Checkpoint 5 · Prototype Ready & User Flows

System Architecture, Screen Mockups, Storyboard & Test Plan | INT335 Design Thinking

1. System Architecture & Prototype Overview

ReturnLoop LPU is structured as a responsive full-stack progressive web application engineered specifically for mobile devices with ■0 infrastructure dependencies. It enables rapid lost/found logging, real-time trait matching, and campus security admin verification.

Figure 1: 6-Panel User Storyboard

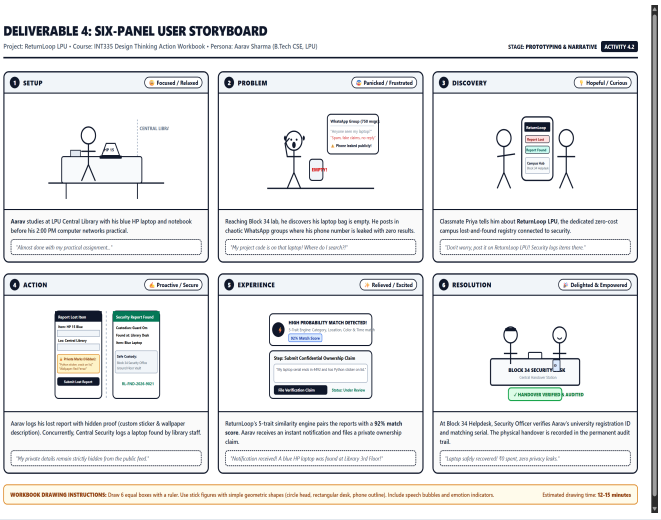


Figure 2: Architecture & User Flow

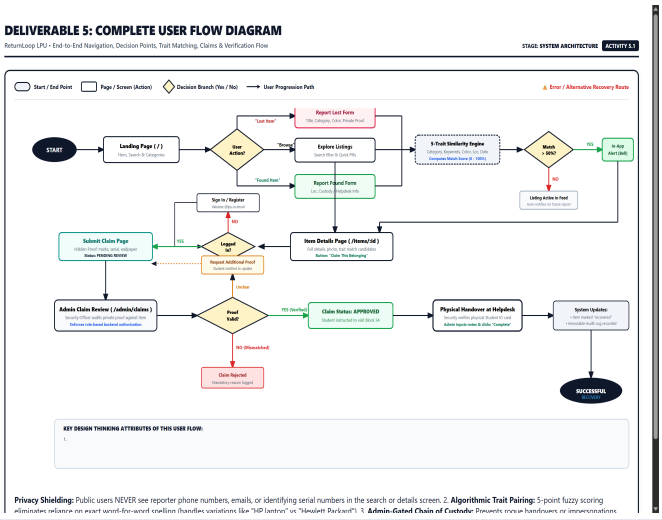
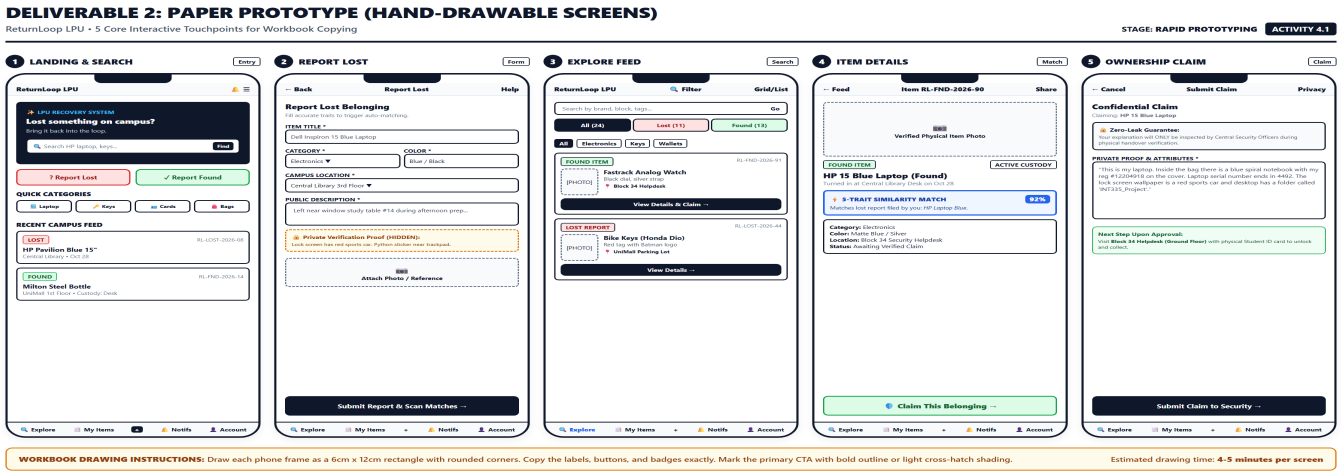


Figure 3: Paper Prototype Screens (Section 13 Compliance)



2. Primary User Scenarios & Prototype Verification Protocol

Scenario #	User Goal & Persona	Step-by-Step Prototype Flow	Success Criteria
Scenario A	**Report Lost Calculator** Aarav (Hosteller)	1. Tap '+ Report' on mobile navbar. 2. Select 'Lost Belonging'. 3. Choose 'Electronics' & input model. 4. Enter secret identifier (scratched initials).	Report completed in < 60s; secret proof hidden from public view.
Scenario B	**Report Found Keys** Priya (Day Scholar)	1. Open quick report modal. 2. Select 'Found Item' & Block 34. 3. Upload photo & set drop status. 4. Receive automated match notification.	Instant notification triggered; no personal phone number exposed.
Scenario C	**Admin Handover** Guard Gurpreet	1. Open Admin Desk portal. 2. Compare owner's submitted proof against physical item. 3. Click 'Verify & Handover'.	Audit trail logged; 0% chance of handover to impostors.